An Investigative Study of Non-Documentary TV Footage as a Data Source for Community Composition and Biodiversity Research

Supplementary Material – Appendix S3: Figures and Tables

Christopher Crowder[1,2], Thabisa McAnyana[3], Geoffrey Cook[1]

[1]Department of Biology, University of Central Florida

[2]Wildlife Division, Nebraska Game and Parks Commission

[3]Biology Department, Nebraska Wesleyan University

Corresponding Author:
Christopher Crowder – CrowderFishScience[at]gmail.com

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Figures

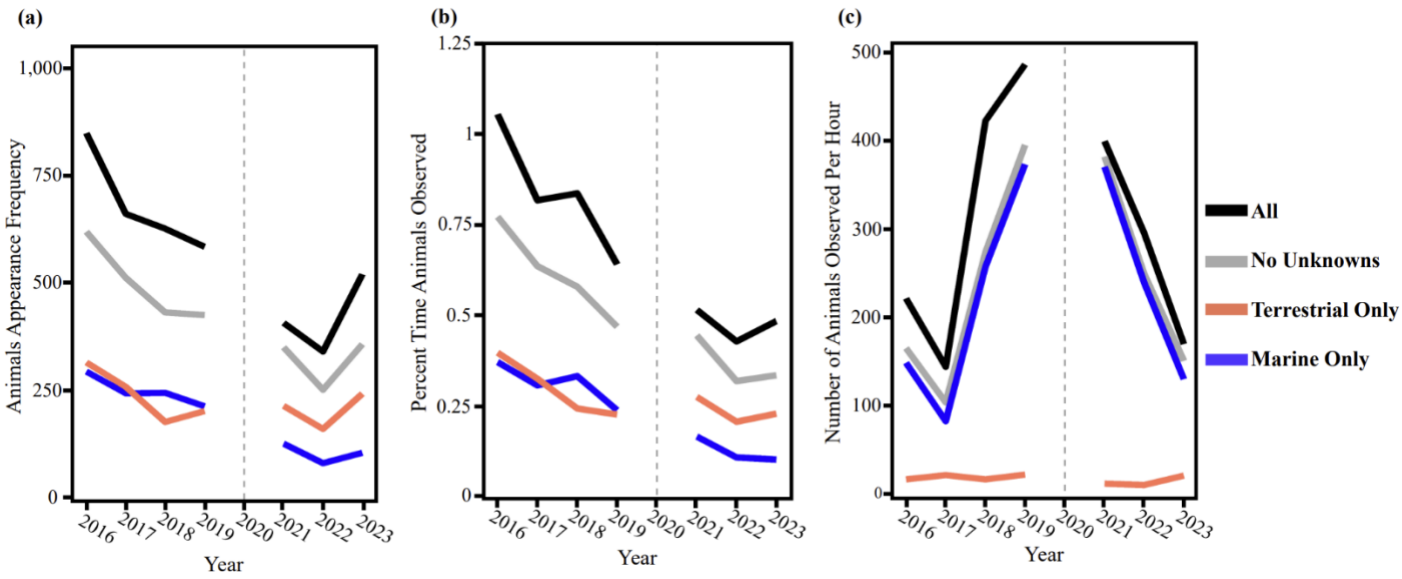


Figure 1. (a) The frequency in which animals appeared on the screen for both seasons filmed within the same year. This is count of appearances, not the abundance of the animals on the screen. (b) The percentage of screen time animals were observed for both seasons in the same year. (c) The mean number of animals observed per one hour of screen time. Please note the omission of 2020 in the figures. This was done to not obscure the trends in the way they were viewed.

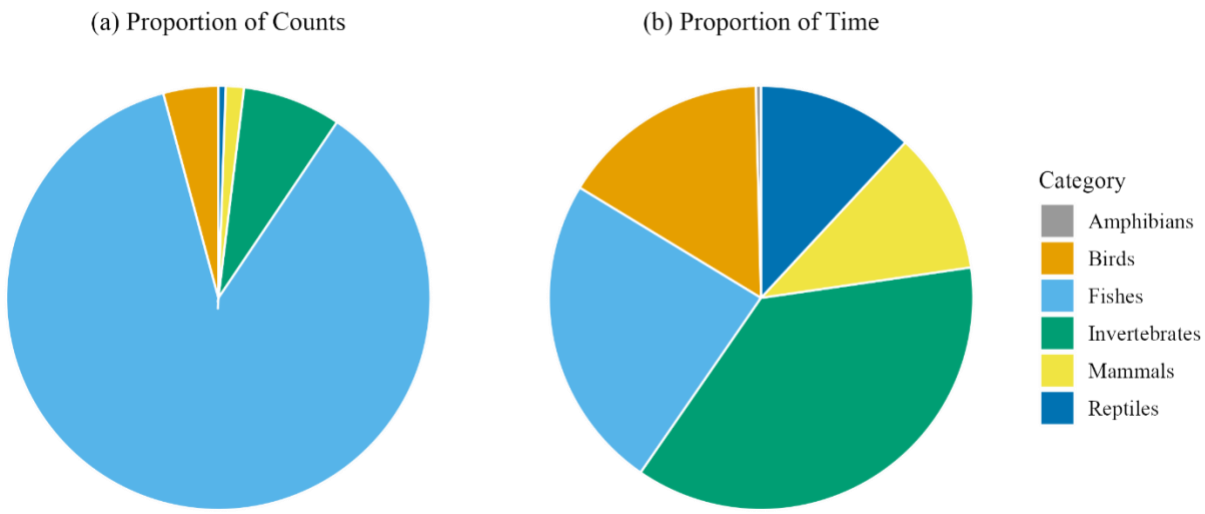


Figure 2. Representation by broad animal category of (a) the proportion of total individual abundance and (b) the proportion of total screen time attributed to each group. Fishes dominated total abundance, whereas invertebrates and fishes accounted for the greatest proportion of screen time.

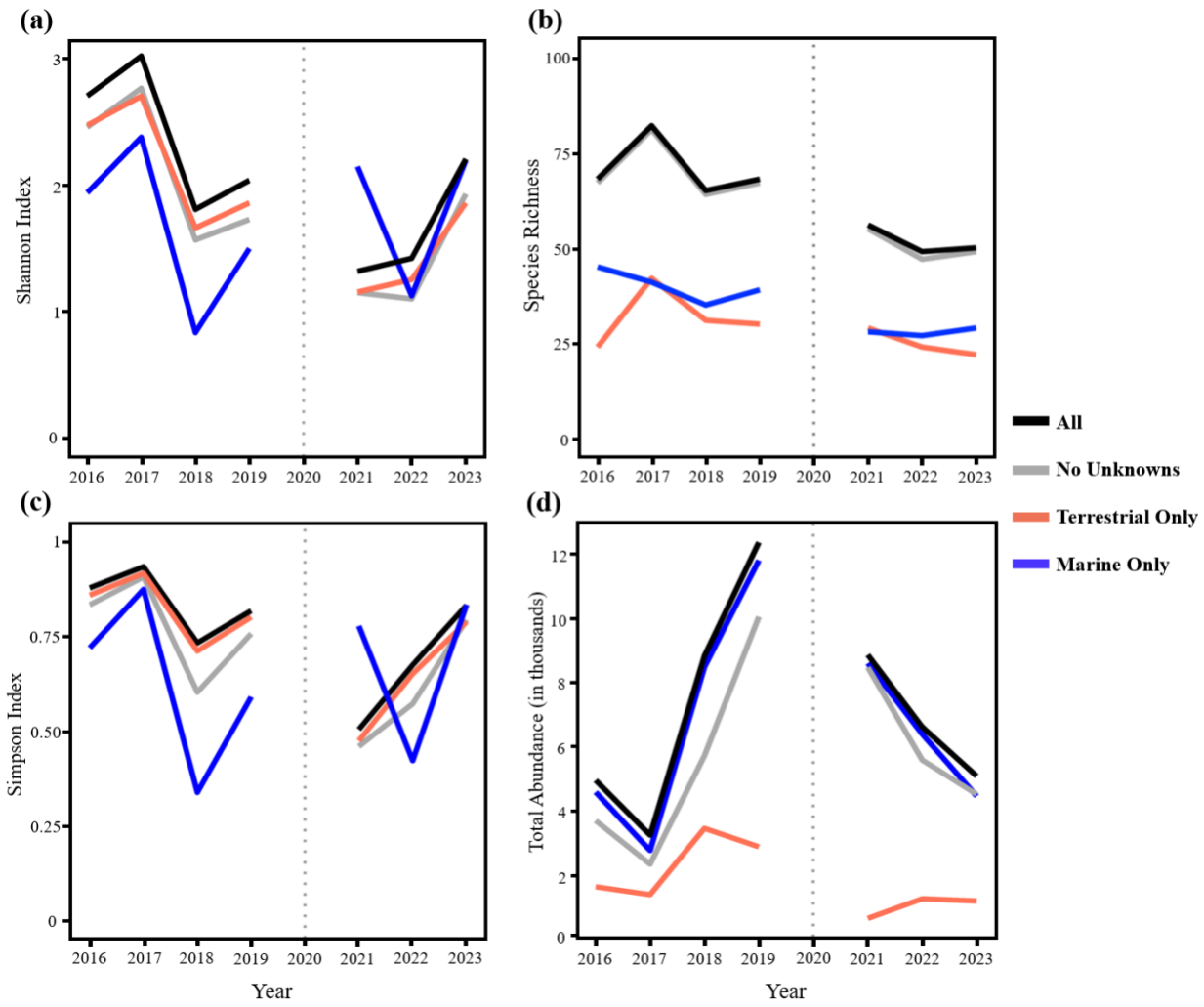
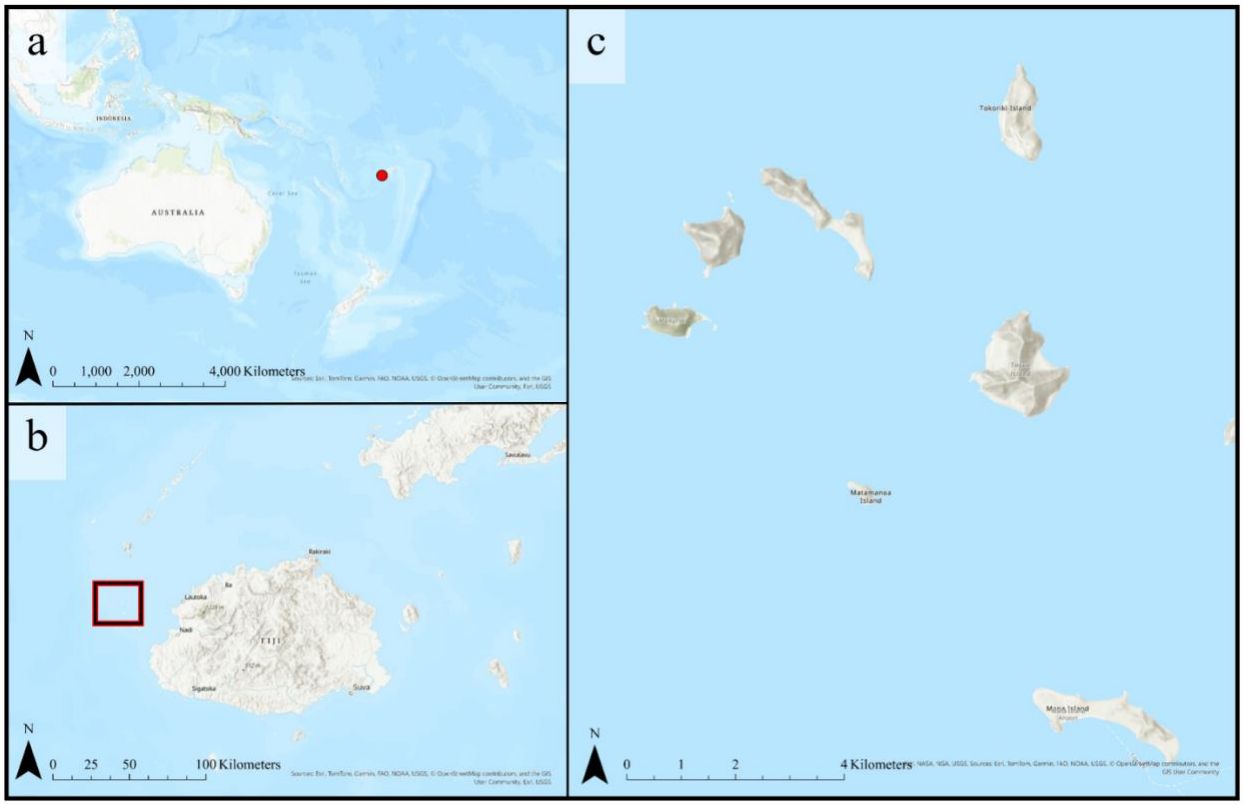
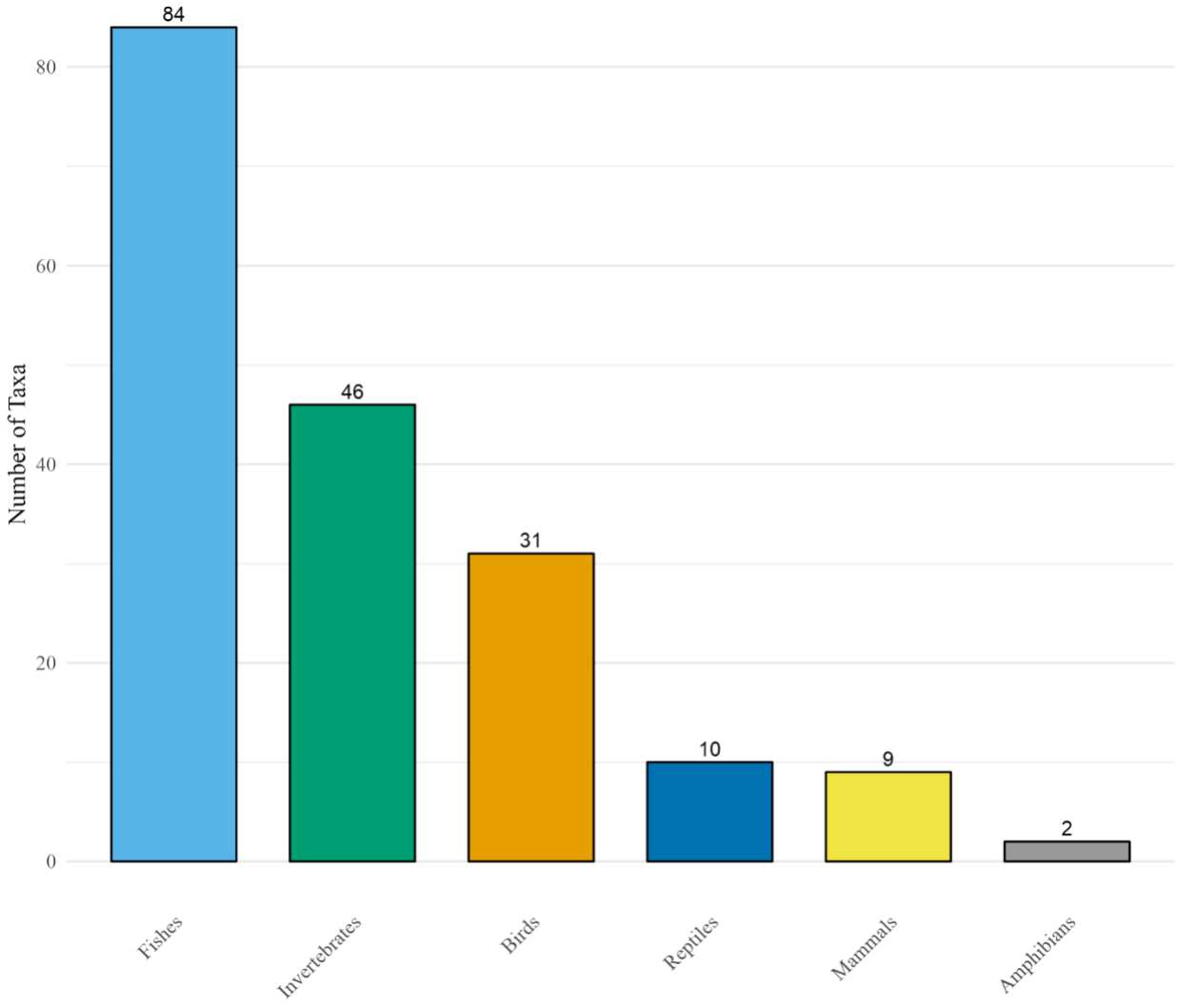


Figure 3. (a) Shannon index, (b) species richness, (c) Simpson diversity index, and (d) total abundance for all animals, all but unknown animals, terrestrial animals only, and marine animals only. Grouping these separately provides insight into variation in these metrics based on the environment and identification of these animals. Please note the omission of 2020 in the figures.

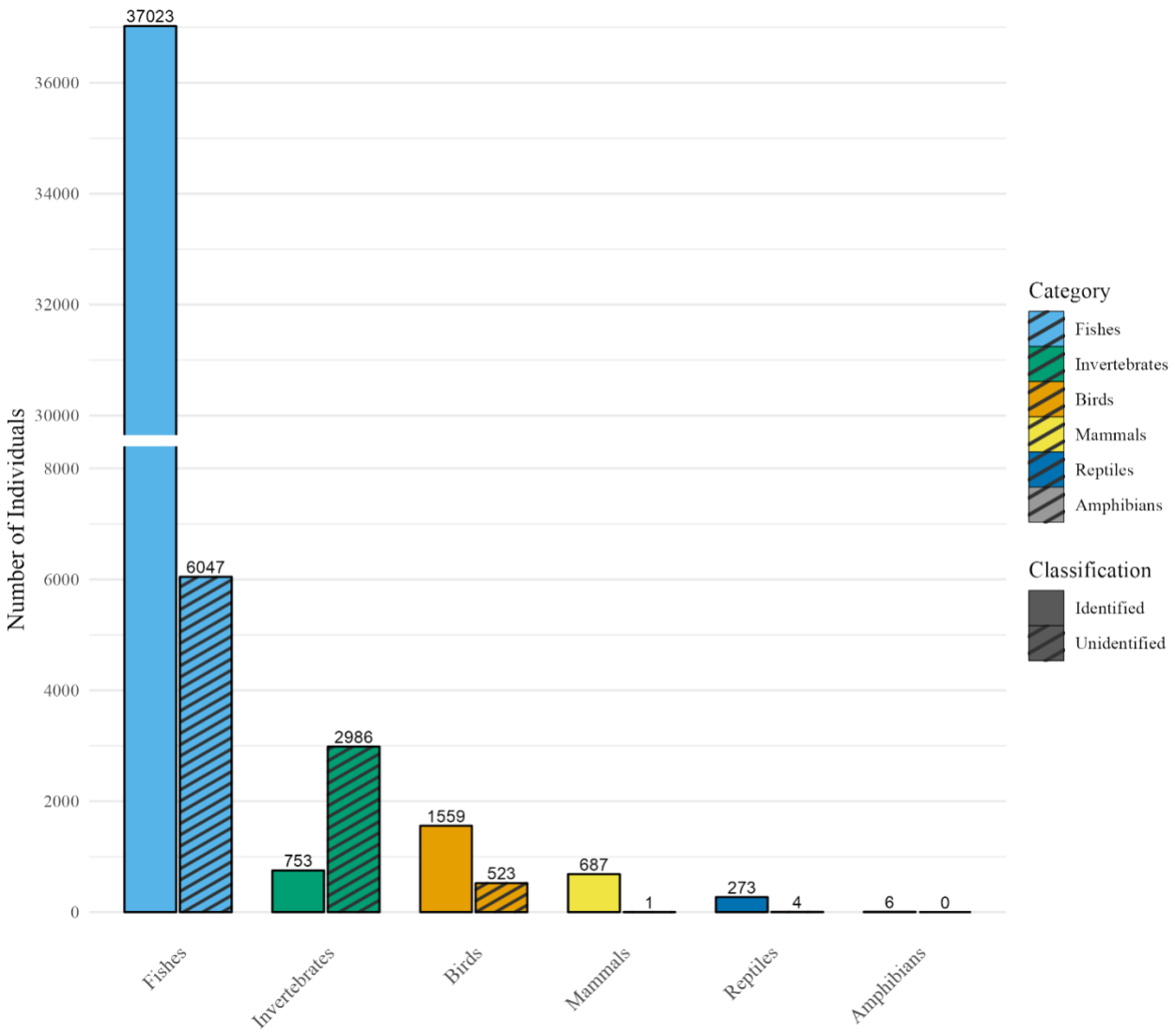
Supplemental Figures



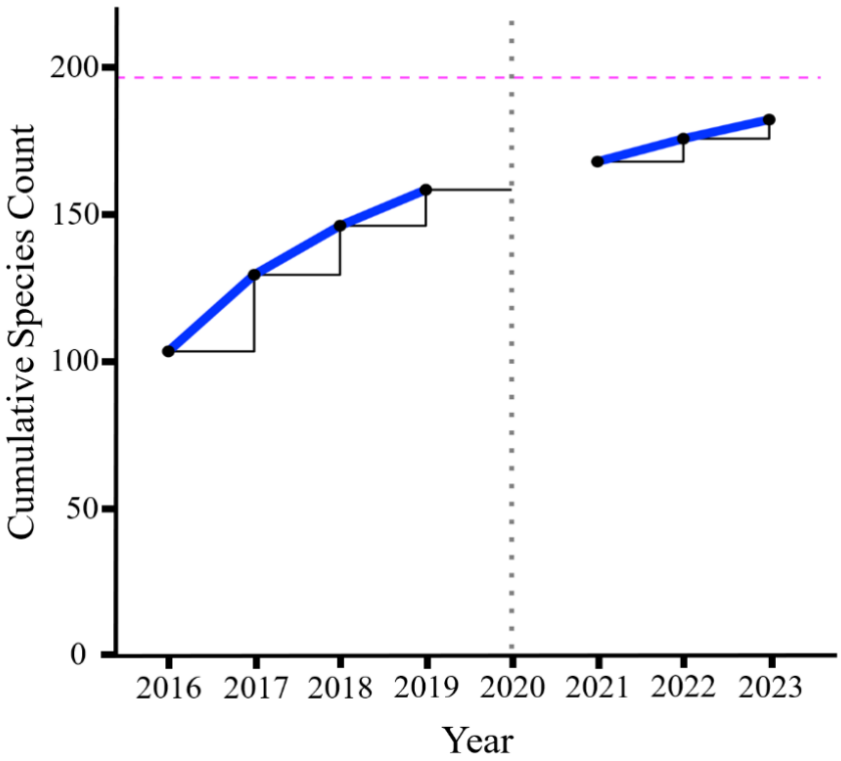
Supplemental Figure 1. (a) Map of Pacific Ocean/Oceania region with Fiji within black box. (b) The Mamanuca Islands, within the red and black box, are off the west coast of Viti Levu. (c) The most common filming locations within the Mananuca Islands, at approximately 17.673° S , 177.107° E.



Supplemental Figure 2. Number of taxa identified within each broad animal category.



Supplemental Figure 3. Number of individuals counted within each broad animal category separated by those that could be identified to the family, genus, or species level and not that could not (i.e., "Unidentified"). Please note the break in the y-axis.



Supplemental Figure 4. Accumulation curve for cumulative taxa counts for each year of filming for those that could be identified down to the family, genus, or species level. Black line is cumulative taxon count displayed stepwise while the blue line is linear connecting each point by year. The pink line represents the theoretical asymptote of the estimated maximum number of unique taxa from this data source (n = 198) if editing and footage collection methods continue to be similar. Please note the omission of 2020 in the figure.

Tables

Table 1. Counts of all individual animals identified down to genus or species level. Individuals that could not be identified to the genus or species level are enumerated in the "count unidentified" column. The amount of time (in seconds) for each screen classification category and sub-category is provided for each animal by the type of footage captured.

Animal Category	Number of Taxa Identified	Count Identified	Count Unidentified	Total Screen Time (sec)	Transition Time (sec)	Interaction Time (sec)	Hunting Time (sec)	Background of Focus Time (sec)	Theme Song Time (sec)
Amphibians	**2**	**6**		**15**	**15**				
Frogs	1	2		3	3				
Toads	1	4		12	12				
Birds	**31**	**1,559**	**523**	**640**	**595**			**42**	**3**
Birds of Prey	2	10		21	21				
Foraging or Shore Birds	29	1,549	523	619	574			42	3
Fishes	**84**	**37,023**	**6,047**	**968**	**856**	**29**	**67**	**6**	**10**
Rays	2	18		49	37		10		2
Reef Fishes	76	36,857	6039	738	653	29	44	6	6
Sharks	6	148	8	181	166		13		2
Invertebrates	**46**	**753**	**2,986**	**1,481**	**585**	**159**	**122**	**615**	
Ants			1,431	92	68	12	2	10	
Bees/Wasps	1	304	238	18	17			1	
Beetles			3	7	5	2			
Butterflies/Moths	6	9	51	83	21	11		51	
Crabs	9	261	56	263	230	12	21		
Flies/Gnats/Dragonfly	2	61	434	143	30	34		79	
Jellyfish	1	2		5	5				
Lobsters	1	2		6	3		3		
Octopus/Cuttlefish	2	11		66	34		32		
Other Bugs	10	34	3	132	48	78	5	1	
Sea Stars	1	2		3	2		1		
Shellfish or Barnacles	1	2	24	22			22		
Spiders	11	62	6	121	117	1		3	
Unknown Flying			739	483	4	9		470	
Worms	1	3	1	37	1		36		
Mammals	**9**	**687**	**1**	**433**	**392**	**2**	**36**	**1**	**2**
Bats	1	557		259	256			1	2
Dolphins	2	69	1	48	48				
Goats	1	21		44	8		36		
Rats	1	15		24	22	2			
Whales	4	25		58	58				
Reptiles	**10**	**273**	**4**	**480**	**450**	**25**		**4**	**1**
Lizards	7	79	2	147	141	2		4	
Snakes	2	151	2	305	295	10			
Turtles	1	43		28	14	13			1
TOTAL	**182**	**40,301**	**9,561**	**4,017**	**2,893**	**215**	**225**	**668**	**16**

Table 2. Top five most abundant taxa identified to the genus level for each of the broader animal groupings. Where possible, taxonomically related species were grouped at the genus or family level (e.g. terns).

Amphibians			**Birds**			**Fish**		
Common Name	Scientific Name	Count	Common Name	Scientific Name	Count	Common Name	Scientific Name	Count
Cane Toad	*Rhinella marina*	4	Wedge-tailed Shearwater	*Ardenna pacifica*	864	Fiji Sardinella	*Sardinella fijiensis*	17,080
Fiji Tree Frog	*Platymantis vitiensis*	2	All Terns	*Thalasseus bergii/Sterna sumatrana/Gygis alba/ Onychoprion fuscatus*	248	Anchovies	*Encrasicholina spp.*	7,641
	-		All Boobies	*Sula sula/ Sula leucogaster*	139	Indian Mackerel	*Rastrelliger kanagurta*	5,648
	-		Black Noddy	*Anous minutus*	117	Snappers and Fusiliers	*Caesio spp./Pterocaesio spp./Lutjanus spp./ Macolor spp.*	3,594
	-		Pacific Reef Heron	*Egretta sacra*	101	Chromis	*Chromis spp.*	766
Invertebrates			**Mammals**			**Reptiles**		
Oriental Paper Wasp	*Polistes olivaceus*	304	Flying Fox	*Pteropus spp.*	557	Pacific Tree Boa	*Candoia bibroni*	136
Tawny Hermit Crab	*Coenobita rugosus*	83	Spinner Dolphin	*Stenella longirostris*	52	Fijian Iguana	*Brachylophus spp.*	55
Smooth-handed Ghost Crab	*Ocypode cordimanus*	73	Goat	*Capra aegragrus hircus*	21	Hawksbill Sea Turtle	*Eretmochelys imbricata*	43
Mottled Lightfoot Crab	*Grapsus albolineatus*	65	Bottlenose Dolphin	*Tursiops spp.*	17	Yellow-lipped Sea Krait	*Laticauda colubrina*	15
Fiji Muscidae Fly	*Dichaetomyia spp.*	34	Polynesian Rat	*Rattus exulans*	15	All Skinks	*Emoia spp.*	12

Table 3. Linear model results for Shannon and Simpson diversity indices, species richness, and abundance. Metrics were grouped by calendar year. Further, the results were grouped by (a) all animals, (b) all but unknown, (c) terrestrial only animals, and (d) marine animals only.

a

Measurement	Intercept	Slope	R^2
Shannon	2.88	-0.19	0.41
Simpson	0.89	-0.03	0.22
Richness	81.57	-4.78	0.76
Abundance	6102.10	255.30	0.03

b

Measurement	Intercept	Slope	R^2
Shannon	2.88	-0.19	0.41
Simpson	0.89	-0.03	0.22
Richness	81.56	-4.78	0.76
Abundance	4052.90	423.4	0.11

c

Measurement	Intercept	Slope	R^2
Shannon	2.88	-0.19	0.41
Simpson	0.89	-0.03	0.22
Richness	42.12	-3.17	0.82
Abundance	3643.00	417.10	0.11

d

Measurement	Intercept	Slope	R^2
Shannon	2.88	-0.19	0.41
Simpson	0.89	-0.03	0.22
Richness	50.28	-1.67	0.35
Abundance	706.01	39.79	0.08

Supplemental Tables

Supplemental Table 1. The season numbers, cumulative run time of the show, number of animals observed, and the total amount of time animals were observed on screen by filming year. The cumulative total across all years is presented in the final row.

Filming Year	Season #	Total Run Time (hrs)	Total Number of Animals Observed	Time Animals on Screen (sec)
2016	33/34	22.30	4,946	850
2017	35/36	22.45	3,236	664
2018	37/38	20.83	8,811	630
2019	39/40	25.35	12,349	588
2021	41/42	22.11	8,845	413
2022	43/44	22.32	6,602	347
2023	45/46	29.87	5,073	525
7 years	**14 seasons**	**165.23**	**49,862**	**4,017**